

Video Game Programming: Introduction

L01

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Icebreaker

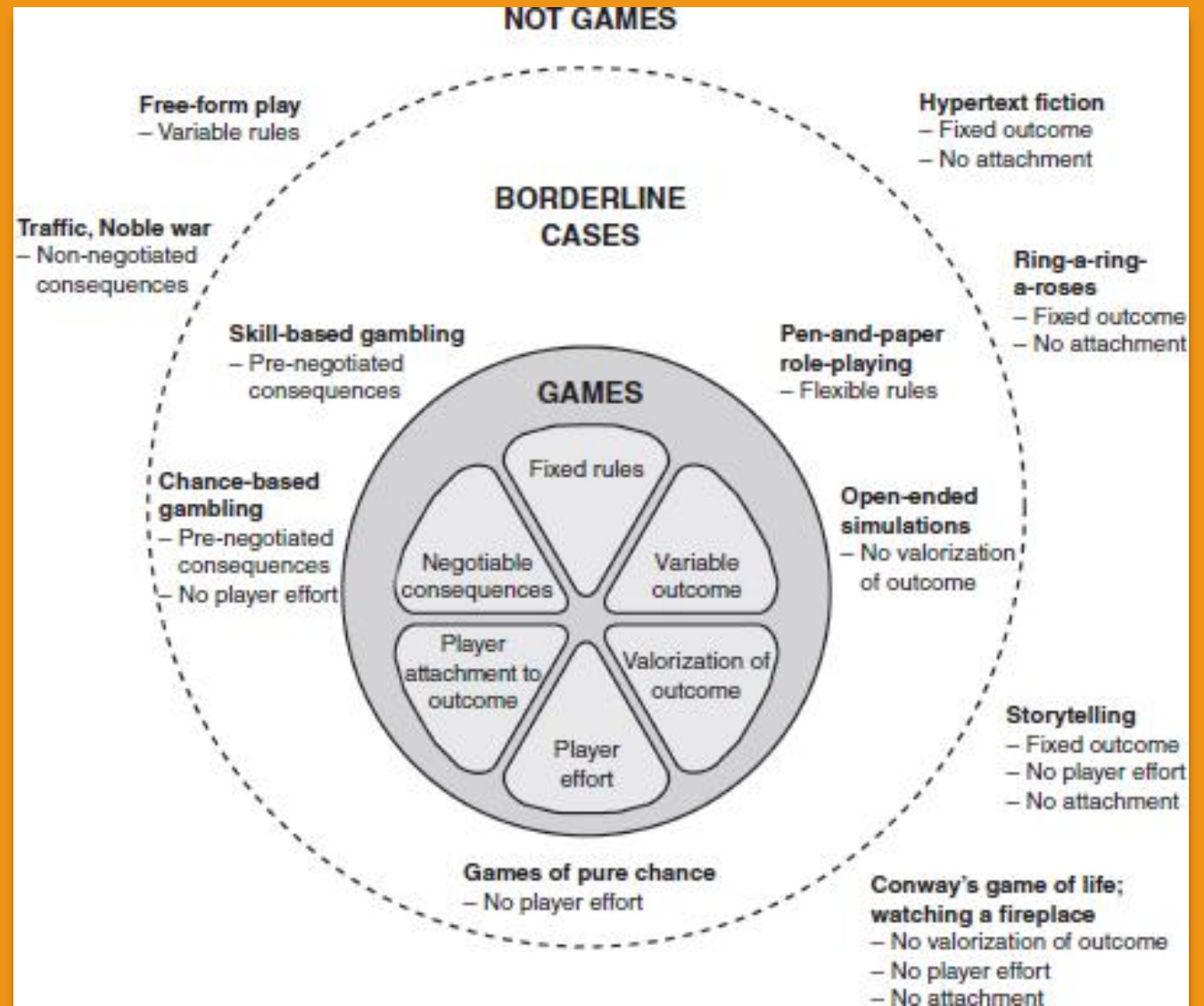
- Name of at least three video games for different platforms
- A facilitator game to help members of a team to know each other
 - Do you play video games? How much?
 - Have you ever programmed a video game?
 - Are you familiar with Unity game engine?
 - Which degree course are you enrolled in?
 - The little-known fact (about video games)
 - Online survey: <https://forms.office.com/e/N2PTYAEuHx>

What is a game?

- Game scholars do not agree on this most fundamental issue (Egenfeldt-Nielsen, Smith, Tosca, Understanding Video Games, 2024)
- A game is a series of interesting choices (Meier)
- A game is a system in which players engage in an artificial conflict, defined by rules, that results in a quantifiable outcome (Salen, Zimmerman, Rules of Play, 2004)
- A game is a rule-based formal system with a variable and quantifiable outcome, where different outcomes are assigned different values, the player exerts effort in order to influence the outcome, the player feels attached to the outcome, and the consequences of the activity are optional and negotiable (Juul, The game, the player, the world: Looking for a heart of gameness, Level Up—Digital Games Research Conference, 2003)

What is (not) a game?

- Juul, Half-real: Video games between real rules and fictional worlds, PhD thesis, IT University of Copenhagen, 2003



Pre-History of Video Games

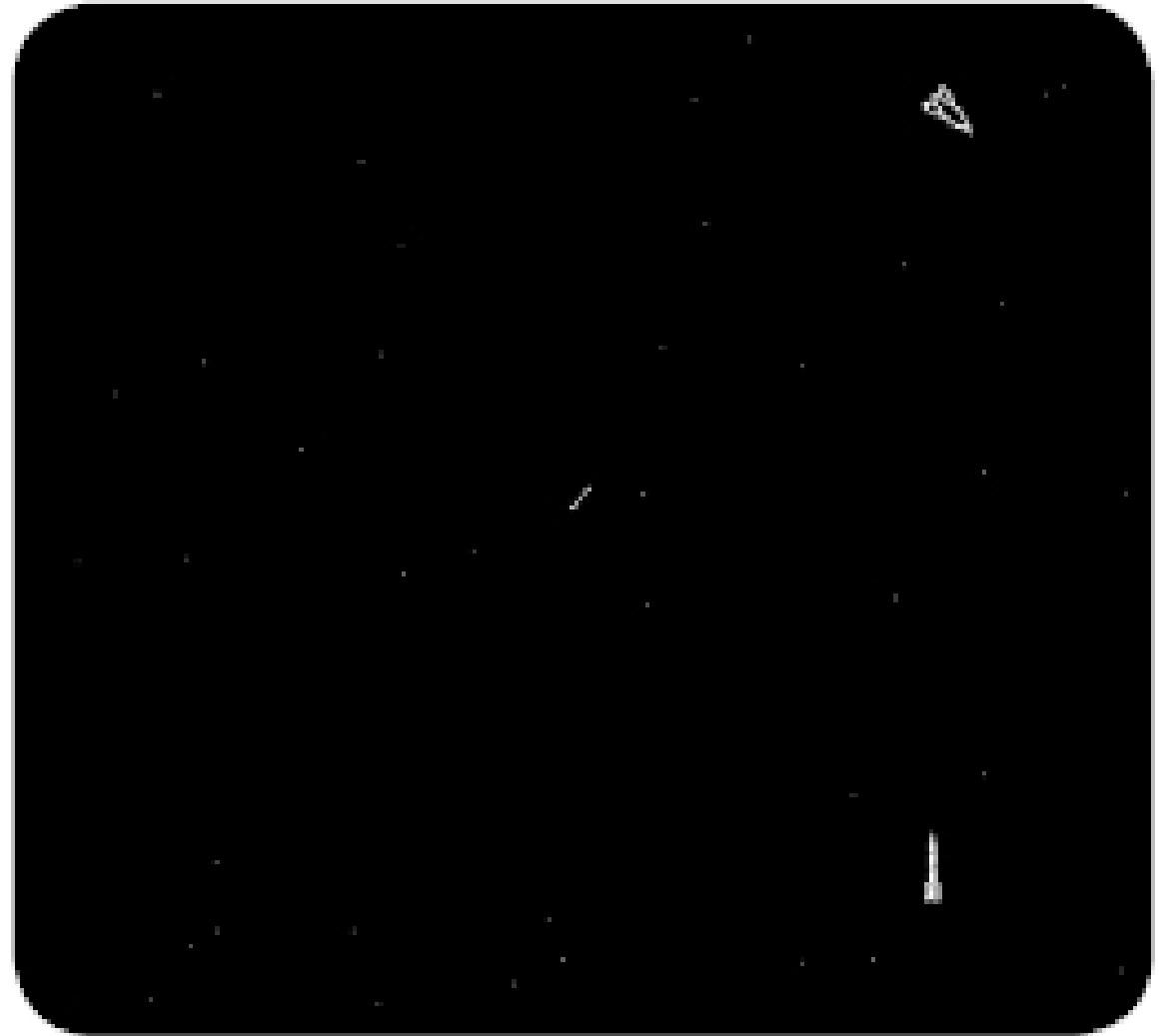
- Senet, Egypt, 2600 BC, game of skill and chance not unlike present-day backgammon
- Go, China, 400 BC, strategy board game
- Monopoly, 1930, board game
- Dungeons & dragons, 1974, fantasy role-playing game
- Cyberpunk, 1988, science-fiction role-playing game



CD Projekt, 2020

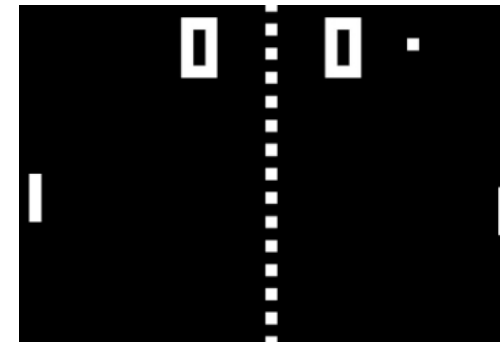
Early history of video games

- Noughts and Crosses, 1952, computerized version of tic-tac-toe, by Douglas (PhD student), University of Cambridge
- Tennis For Two, 1958, precursor of Pong, by Higinbotham, Brookhaven National Laboratory
- Spacewar!, 1962, science-fiction war game, MIT (in fig.)



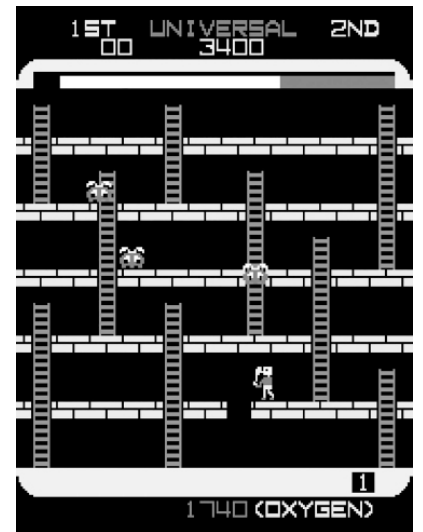
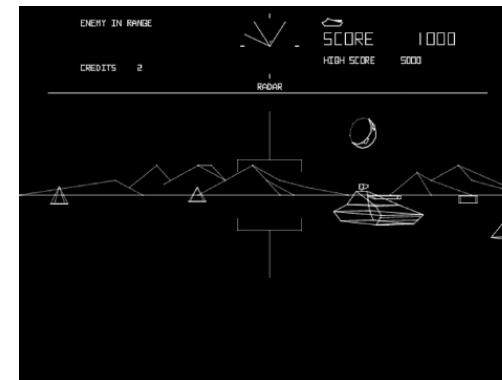
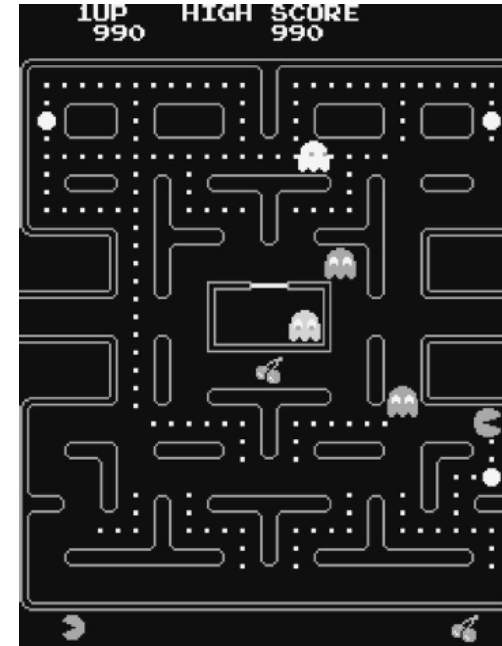
Game industry in the '70s

- Arcade game: game played on dedicated “arcade” machines by inserting coins
- Pong (arcade), 1972, by Atari (in figures)
- Hunt the Wumpus and Adventure, 1972, text adventures
- Channel F, 1976, first console with cartridges
- Night Driver, 1976, first in first-person (in figure)
- Death Race, 1976, first controversial
- Space Invaders, 1978 (in figure)



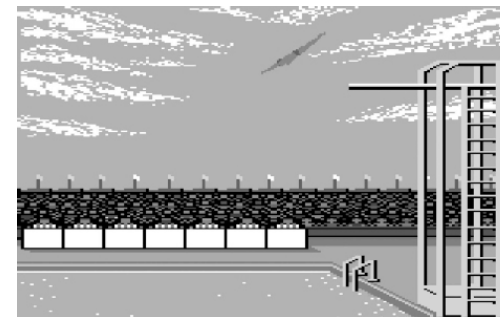
The '80s until Crash

- Pac-Man (arcade), 1980 (in fig.)
- Battlezone, 1980, tank fighting with geometric shapes, realistic, used for training (in fig.)
- Space Panic, 1980, first platform game (in fig.)
- Donkey Kong, 1981, first with Mario
- Pole Position, 1982, first-person driving game
- Zaxxon, 1982, first with an isometric (2.5D) perspective (in fig.)
- Mario Bros., 1983, with cooperation
- Great Video Game Crash, 1983, due to poor games for Atari 2600 and proliferation of PCs



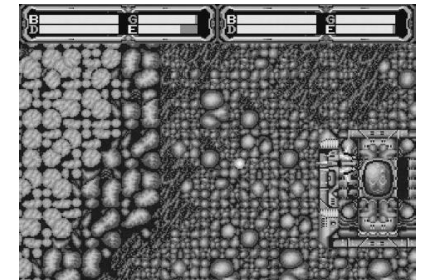
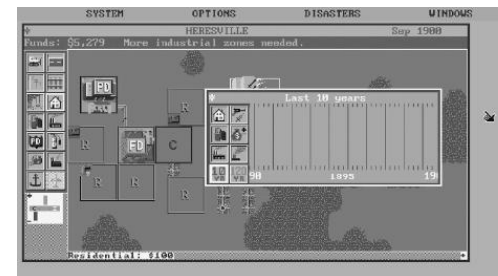
From Crash to NES

- Rise of Apple and Commodore
- Deadline, 1983, interactive fiction, solve the murder (suicide?)
- King's Quest, 1983, text input with graphic output (in fig.)
- Summer Games (II), 1984, new genre of sport games (in fig.)
- Elite, 1984, one of the first home video games with polygon 3D graphics (in fig.)
- Gauntlet, 1985, up to 4 co-op players with different roles and abilities, innovative role-playing (in fig.)
- Nintendo Entertainment System (NES) in the United States, 1986



'80s after NES

- Microsoft Flight Simulator, 1986, new standard for realism of simulations (in fig.)
- Maniac Mansion, 1987, point-and-click instead of text adventure (in fig.)
- Final Fantasy, 1987, RPG series
- Prince of Persia, 1989, new jump timing
- GameBoy and Lynx, 1989, handheld consoles
- Herzog Zwei, 1989, combines strategy and action without turns (in fig.)
- Populous from 1989, combined real-time and isometric perspective, then standard
- SimCity from 1989, player assumed the role of mayor and managed a city (in fig.)



Strategy & FPS in the '90s

- Civilization, 1991, turn-based strategy series
- Dune II, 1992, real-time strategy (in fig.)
- Warcraft, 1994, Command & Conquer, 1995, Age of Empires, 1997, and StarCraft, 1998, followed
- Multiplayer using LAN
- World Wide Web allow multiplayer expansion
- Ultima Online, 1997, with monthly subscription
- Wolfenstein 3D, 1992, first first-person shooter (in fig.)
- Doom, 1993, allowed for mods (in fig.)
- Quake, Unreal, Half-Life followed



Adventures, platform, and ratings in the '90s

- Mortal Kombat, 1992, extreme violence led to Videogame Rating Act in 1994
- The Secret of Monkey Island, 1990, a pirate parody adventure
- Sonic the Hedgehog, 1991, platform series
- Myst, 1993, strong narrative, fixed choice
- Baldur's Gate, 1998, large freedom of choice within the game, e.g., whether and how to complete quests and engage with subplots
- Tomb Raider, 1996, combines platform, action, and adventure
- The 7th Guest, 1993, first to use CD-ROM
- Phantasmagoria, 1995, with real actors used 7 CDs
- Final Fantasy VII, 1997, first 3D game of the series
- 3D polygon graphics gradually replaces 2D
- Well-equipped PCs vs. Playstation, Nintendo 64, Dreamcast

The new millennium

- Popularity of horror series (Resident Evil, Silent Hill) and multiplayer shooters (Unreal Tournament, Quake III: Arena) released in 1999
- Diffusion of MMORPG like World of Warcraft, 2004, and Final Fantasy XI, 2003
- Halo, 2001, and new FPS with more interactive objects (Max Payne II, Far Cry)
- Even higher interaction and lots of controversy with GTA III
- New series such as BioShock, Assassin Creed, Uncharted
- User generated content in games like Spore, 2008
- Console war: PS2, PS3, Xbox, Xbox 360, Wii
- Casual games popularity: Wii, Dance Dance Revolution, Guitar Hero, Rock Band, browser, mobile
- Release of iPhone, 2007, allows indie developers to sell thousands of copies
- Games reach (back) universities with Digital Games Research Association (DiGRA) conference and dedicated scientific journals

Recent years

- New digital distribution on mobile, consoles, and PCs (e.g., app stores)
- Expansion of downloadable content (DLC) and in-game purchase
- Indie can become mainstream (e.g., Limbo, Flow, Journey, Minecraft)
- Mobile successes by big producers as well as independent developers (Assassin's Creed Identity, Angry Birds, Candy Crush)
- Social games, e.g., FarmVille, 2009, and similar, criticized, but popular
- Rise of immersive home VR (e.g., Oculus, Vive, PS VR)
- Natural interfaces using computer vision (e.g., Kinect) and VR trackers
- Context awareness and AR (Ingress, Pokemon Go)
- Game elements in non-game activities (i.e., gamification)

Questions and lessons from history

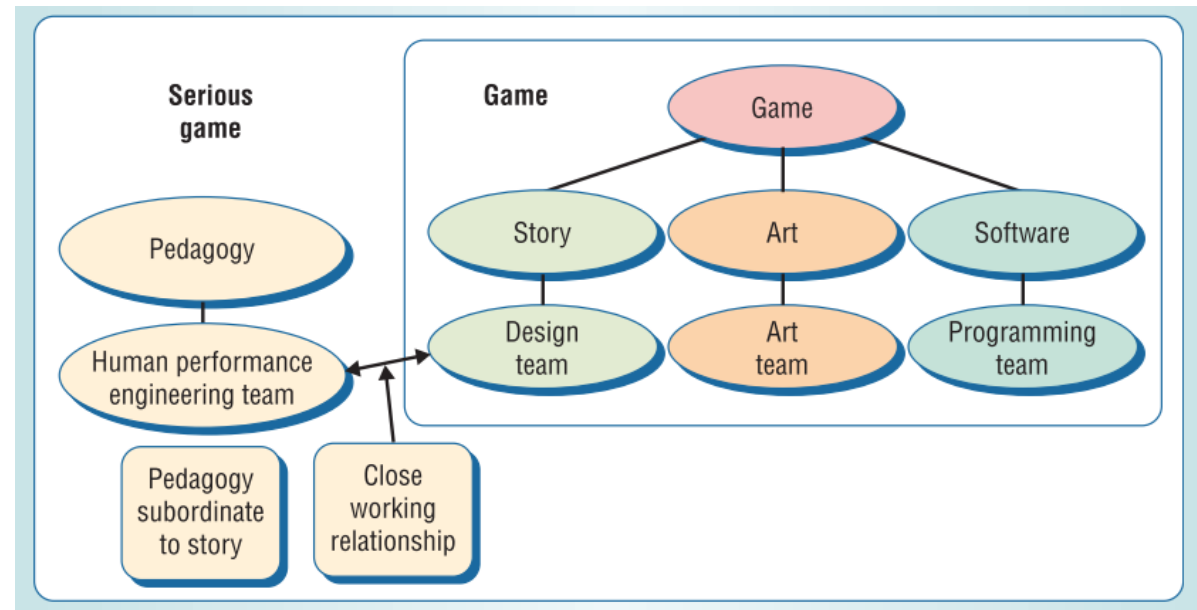
- Games like movies? Multiplayer vs. single-player games? Natural interfaces vs. controllers? AAA games teams vs. indie creativity?
- Game history shows several unpredicted events that changed the field
- Most (not all) first-of-their kind games and games that exploit current technologies became (sometimes unexpectedly) popular
- The ability to maneuver in this changing world, to evaluate new trends, and ultimately to design worthwhile and innovative games depends on understanding the history of video games (Egenfeldt-Nielsen, Smith, Tosca, Understanding Video Games, 2024)
- Research on games can propose solutions for future trends and show how to apply game lessons to non-game domains

What is a video game?

- Video game: a mental contest, played with a computer according to certain rules for amusement, recreation, or winning a stake (Zyda, Computer, 2005)
- Serious game: a mental contest, played with a computer in accordance with specific rules, that uses entertainment to further government or corporate training, education, health, public policy, and strategic communication objectives (Zyda, Computer, 2005)

Who makes a game?

- One person does all
- Distinct (small or large) teams for different aspects
- Interesting schema for (serious) games by Zyda (Computer, 2005)
- This course focuses on programming team skills and (partly) design team skills



Game career options

- Game programmers have three main career options:
 - Game programmer in a company
 - Independent game programmer
 - Academic research on games
- For all the options, tasks can involve programming of
 - Entertainment games
 - Serious games

Game programmer in a company

- Different sites list entertainment game companies in Central EU:
 - Gamecompanies.com: about 25 game companies in Italy, 5 in Slovenia, 15 in Austria, and 100 in Germany
 - Gameprog.it: about 100 Italian game companies
 - Slovenia Game Industry: 24 game companies
- Some are very small, others have hundreds of employees
- Some collaborate with big studios and (small or big) publishers
- Overall, they cover all game platforms (PC, console, XR, Web, mobile)
- If companies working on serious games and gamification are included, the numbers are increasingly growing
- Some former students at University of Udine work or worked in these companies or for companies outside EU

Independent game programmer

- Some former students created their own companies that develop (serious) games or became independent game developers
- Different kinds of supports to start an independent game project:
 - Interreg ITA-SLO DIVA project
 - Italian Ministry of Economic Development First Playable Fund
 - Crowdfunding
- Different ways to get noticed:
 - Game jams (e.g., Ludum Dare)
 - Start cup or software competitions

Academic research on games

- Some former students choose the academic career working on (serious) games as PhD students and/or research fellows
- Game academic research includes:
 - Designing, developing, and evaluating serious games
 - Studying players' experience, engagement, behavior, physiological parameters,...
 - Evaluating new input and output game devices
 - Designing and evaluating game locomotion techniques
 - ...

Journals publishing game research

- Specific journals:
 - Entert. Computing (Elsevier)
 - Trans. on Games (IEEE)
 - Computers in Entert.(ACM)
 - The Computer Games J. (Springer)
 - Games for Health J. (Mary Ann Liebert)
 - Simulation & Gaming, Games and Culture (SAGE)
 - Game Studies
- Top-ranked journals publishing paper about games:
 - Trans. Visualization and Computer Graphics (IEEE)
 - Int. J. Human-Computer Studies, Computers & Education, Computers in Human Behavior (Elsevier)
 - Trans. Computer-Human Interaction (ACM)
 - Virtual Reality (Springer)

Topics covered in the course

- Fundamentals of game design to guide programming
- Fundamentals of scripting in a game engine
- Camera, lighting, and audio control
- Project and asset management
- Principles of physics and animation
- Design and implementation of the user interface
- Game architecture, logic, and artificial intelligence
- Design and optimizations for mobile, XR, and online game programming
- Game testing and release (including game analytics)
- Training and vehiculation of serious messages using video games
- Game ethics

General approach

- First part of the lesson: concepts and theories (with slides)
- Second part of the lesson: numerous practical examples using Unity game engine (hands on, without slides)
- In both first and second part: several case studies, including both video games for entertainment and serious games (with and without slides)

How to pass the exam

- Design and develop a complete minigame or at least three game levels (plus menu and tutorial)
- Discuss the project at the exam
- Answer questions about theoretical aspects of the course
- Project evaluation is based on Unity Curricular Framework (see table)

	Exceeds Expectations	Meets Expectations	Below Expectations
Technical Proficiency	Project is technically complete, without bugs or errors Extra project elements are implemented, or student has crafted project elements particularly well Student has achieved learning objectives	Project compiles as expected Project elements are present Project achieves learning objective	Project does not compile or has fatal errors Project cannot be played to completion Project does not demonstrate goal or learning objective at all
Technical Familiarity	Student demonstrates a fluid understanding of their process and results Student can identify and offer a plan for resolving bugs in their code or their output	Student demonstrates a basic understanding of their development process Student can identify but not resolve the source of bugs in their code or output	Student cannot answer questions about their process Student cannot answer questions about their results Student can not identify the source of bugs or flaws in the code or the project
Design Sense	Project demonstrates a clear design sensibility Project displays information in thoughtful or especially concise way Project is easy to use	Project takes users into account Project displays information accurately and clearly Project can be used without student input	Project does not demonstrate an awareness of the user Project is not user friendly or has elements that cannot be accessed or understood without student guidance Project communicates information incoherently

Official texts

- Egenfeldt-Nielsen Simon, Smith Jonas Heide, Pajares Tosca Susana, Understanding Video Games, 5th edition, Routledge, 2024
- Roberts Paul, Artificial Intelligence in Games, CRC Press, 2023
- Gomila Salas Juan Gabriel, Borromeo Nicolas Alejandro, Hands-On Unity Game Development, 4th edition, Packt Publishing, 2024
- All available on <https://udiscover.uniud.it/> as eBooks

Bibliografy

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